

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
6th Semester of B. Pharm. Examination
University Theory Examination October 2015
PH313 Biopharmaceutics and Pharmacokinetics

Date: 19-10-2015 , Monday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any **TWO** of the following

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|---|--|----|
| A | Explain in brief about the types of drug transport mechanisms involved in drug absorption. | 10 |
| B | i) Classify factors affecting drug absorption. | 05 |
| | ii) Explain any two physicochemical factors in detail | 05 |
| C | i) Write a note on volume of distribution and its significance. | 05 |
| | ii) Discuss in brief various types of plasma proteins | 05 |

Q 2 Attempt any **TWO** of the following

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|---|---|----|
| A | Explain various types of compartment models for study of pharmacokinetics. Give its advantages and disadvantages. | 10 |
| B | Explain various pharmacokinetic parameters of one compartment open model for oral administration . | 10 |
| C | i) Describe Wagner Nelson Method for determination of K_a | 05 |
| | ii) Define clearance and elimination half life with equation | 05 |

SECTION- II

Q 3 Attempt any **TWO** of the following

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|---|--|----|
| A | Explain various methods of Bioequivalence study design. | 10 |
| B | What is IVIVC? Explain one method for estimation of IVIVC | 10 |
| C | i) Write the various reasons for non linear pharmacokinetics | 05 |
| | ii) Explain following terminology : Onset of action, Duration of action, AUC | 05 |

Q 4 Attempt any **TWO** of the following

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|----------|--|-----------|
| A | i) Write a note on BCS classes of drugs | 05 |
| | ii) Describe bioequivalence, chemical equivalence and therapeutics equivalence | 05 |
| B | How any drug is excreted from body? Explain mechanism of excretion | 10 |
| C | i) Explain any two methods for enhancement of bioavailability | 05 |
| | ii) Differentiate bioavailability and bioequivalence | 05 |
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